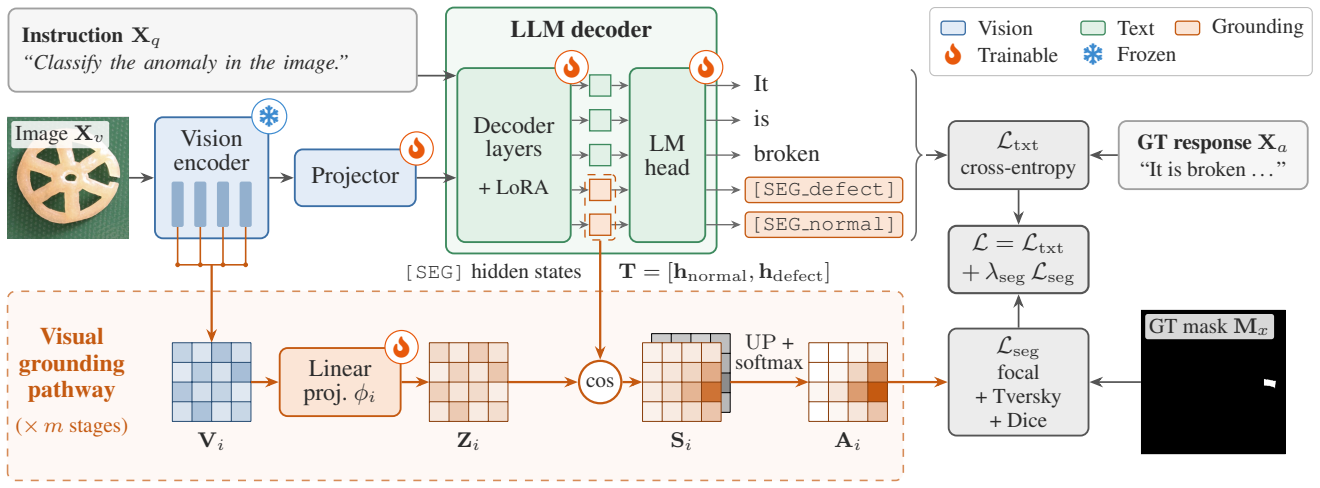

Instruction $\mathbf{X}_q$
"Classify the anomaly in the image."
LLM decoder
Vision
Text
Grounding
Trainable
Frozen
Image $\mathbf{X}_v$
Vision encoder
Projector
Decoder layers
+ LoRA
LM head
It
is
broken
[SEG_defect]
[SEG_normal]
$\mathcal{L}_{\text{txt}}$ cross-entropy
GT response $\mathbf{X}_a$
"It is broken . . ."
[SEG] hidden states
$\mathbf{T} = [\mathbf{h}_{\text{normal}}, \mathbf{h}_{\text{defect}}]$
$\mathcal{L} = \mathcal{L}_{\text{txt}} + \lambda_{\text{seg}} \mathcal{L}_{\text{seg}}$
Visual grounding pathway
(× m stages)
Linear proj. ϕ_i
cos
UP + softmax
$\mathcal{L}_{\text{seg}}$ focal + Tversky + Dice
GT mask $\mathbf{M}_x$
$\mathbf{V}_i$
$\mathbf{Z}_i$
$\mathbf{S}_i$
$\mathbf{A}_i$